



Cloudy Days on Mars?

Curiosity Rover has registered cloudy skies on Mars, an interesting phenomenon for the planet. <http://digit.in/jun21-61>



Huge leap for Quantum Computing

UC Riverside will lead collaborative effort at developing scalable quantum computers. <http://digit.in/jun21-62>

The hunt for Axions

Physicists know these mysterious particles must exist, but haven't been able to spot them

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stronomers have attempted to quantify the amount of mass in a galaxy or galaxy cluster by observing

the movement of stars around the centre of the Milky Way, or galaxies on the fringes of a galaxy cluster, or the edges and rotation of spiral galaxies, and have maintained that these objects have more mass than can be seen. In fact, the stuff that we don't see are more massive than all the stuff that we do see. This dark matter is necessary to explain the movement of the visible matter, and physicists have come up with many hypothetical particles that could constitute dark matter, and one of these are axions. Axions are extremely fast and light particles that barely collide with matter, and pass through most things in the universe as if they were spectres. Axions are considered to be weakly interacting with ordinary matter. Even though they can make up up to 85 percent of the mass of the universe, axions are believed to be a fraction of the mass of a photon, one of the lightest particles known.



The RADES detector at CERN

Axions have been incredibly convenient at explaining some of the most puzzling aspects of science, the maintenance of CP symmetry, where the laws of physics remain the same when a particle is switched out with its corresponding antiparticle, and the spatial coordinates inverted (think of it like a mirror image). Axions explain the mystery of why neutrons do not interact with electric fields despite having a charged component. There also seems to be increasing evidence that axions are, in fact, the dark matter that everyone has been looking to find. Researchers have proposed a mechanism known as "kinetic misalignment", that provides two pieces of good news. One is additional evidence of axion and dark matter equivalence, the second is that it predicts that axions actually

interact with matter more strongly than previously believed, potentially making it easier for scientists to actually spot one. Keisuke Harigaya of the Institute for Advanced Study says, "the extensive literature on the axion was built upon the assumption that the axion field is initially static in the early universe. Instead, we discovered that the axion field may be initially dynamic as a consequence of theories of quantum gravity with axions." The new mechanism allows for far more creation of axions in the early universe than previously predicted.

Physicists predict that if these axions do exist, then they are forged in some of the most extreme and unique conditions in the universe, such as the cores of stars on the verge of going nova. As these stars break out into